PROGRAM CODE:

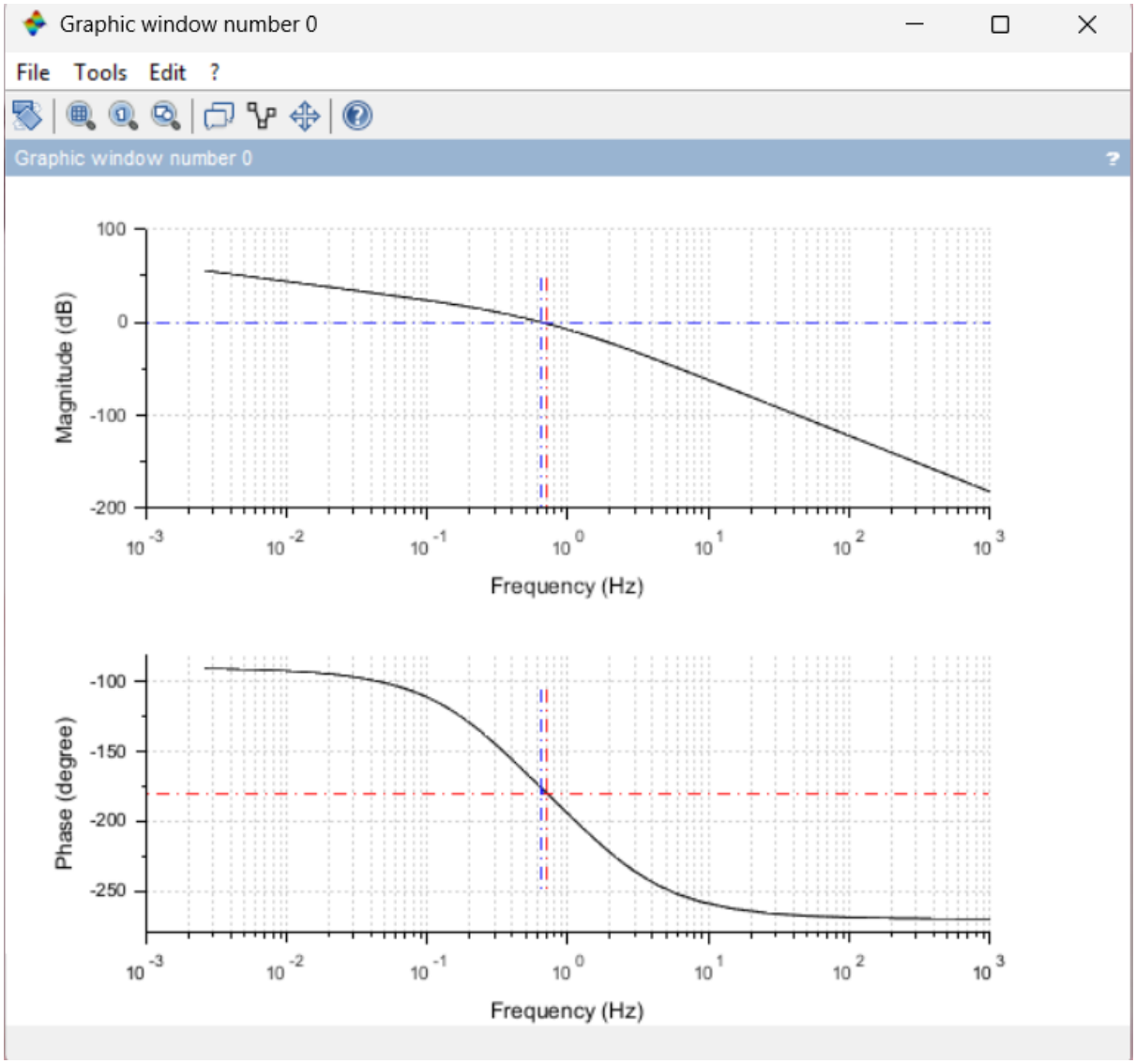
```
clc;
clear;
k=10;
num=poly([1],"s","coeff");
den=poly([0 1 0.6 0.05],"s","coeff");
s1=syslin('c',num,den);
G=k*s1;
disp(G, 'The given transfer function G(s)=');
bode(G,0.01,1000);
xtitle('Bode plot of the given transfer function G(s)');
[g,frp]=g_margin(G);
[p,frg]=p_margin(G)
show_margins(G)
disp(frg,'Gain crossover frequency=',p,'Phase margin(degrees)=');
disp(frp,'Phase crossover frequency=',g,'Gain margin(dB)=');
```

OUTPUT:

Scilab 2025.1.0 Console

```
           10
  -----------------
   s +0.6s^2 +0.05s^3
 "The given transfer function G(s)="
  0.6489779
 "Gain crossover frequency="
  3.9430653
 "Phase margin(degrees)="
  0.7117625
 "Phase crossover frequency="
  1.5836249
 "Gain margin(dB)="

-->
```


Graphic window number 0
File Tools Edit ?
Graphic window number 0
Magnitude (dB)
100
0
-100
-200
10 -3
10 -2
10 -1
10 0
10 1
10 2
10 3
Frequency (Hz)
Phase (degree)
-100
-150
-200
-250
10 -3
10 -2
10 -1
10 0
10 1
10 2
10 3
Frequency (Hz)